INDU 211/2X — Assignment #2 Complete Solutions Guide

Course: INDU 211/2X — Introduction to Production & Manufacturing Systems

Institution: Concordia University · Gina Cody School of Engineering and Computer Science

Department: Department of Mechanical, Industrial and Aerospace Engineering

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Executive Summary & Assignment Structure

This document provides complete, rigorous, and pedagogical solutions for **INDU 211 Assignment #2**, covering:

1. **Problem 1 (True/False & Conceptual Diagnostic):** Comprehensive analysis of 10 core propositions regarding Transportation Linear Programming models, heuristic feasible plans vs. global optimality proofs, sensitivity analysis, and real-world implementation constraints based on the *Plain View Manufacturing Company* model.

2. **Problem 2 (Facility Location & Transportation Problem):** Step-by-step application of the **Lowest Cost First (Least Cost Method)** heuristic to solve a balanced distribution network from two wholesale distribution centers (Louisville and Morgantown) to four regional retail stores (Richmond, Philadelphia, Cleveland, and Knoxville), computing total transportation costs ($2,200,000).

3. **Problem 3 (Plant Intralogistics & Fork Truck Relocation):** Formal Linear Programming (LP) Transportation Model formulation (decision variables, objective function, supply/demand constraints) and execution of the Least Cost Method with dual tie-breaking analysis for relocating 7 fork trucks across 4 supply sites (A, B, C, D) and 3 demand sites (X, Y, Z), resulting in an optimal transfer cost of $25.

4. **Problem 4 (Warehouse Routing & Traveling Salesman Problem):** Formal mathematical formulation of the Traveling Salesman Problem (TSP) with degree and Miller-Tucker-Zemlin (MTZ) subtour elimination constraints, complete execution and distance evaluation of the **Nearest-Neighbor (NN)** heuristic ($28\ \text{units}$ round-trip), comprehensive analysis of the **Next-Nearest-Neighbor (NNN)** heuristic ($38\ \text{units}$ under the literal "each time" rule; $31\ \text{units}$ under the 1st-step variant), and a combinatorial proof identifying the true global optimal tour ($27\ \text{units}$).

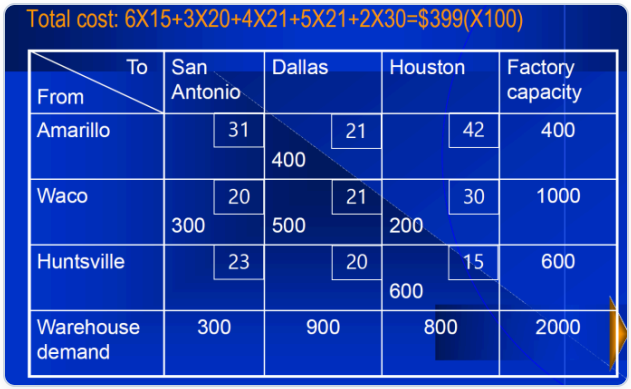
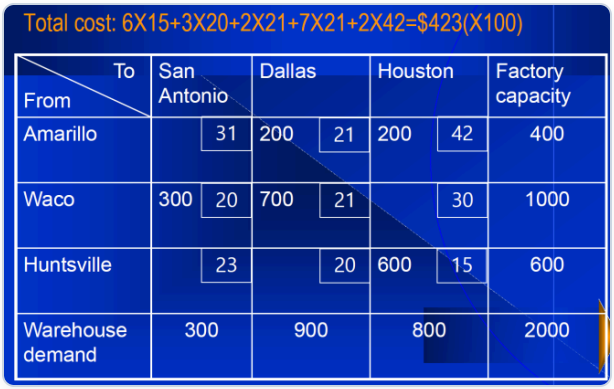
Problem 1: True / False Transportation Model Conceptual Analysis

1.1 Problem Statement & Background Model

The problem presents two feasible transportation plans for the **Plain View Manufacturing Company** distribution network (introduced in Lecture 4, Slides 10–16):

- **Factories (Sources):** Amarillo (capacity 400), Waco (capacity 1,000), Huntsville (capacity 600). Total factory capacity = $2,000\ \text{units/month}$.

- **Warehouses (Destinations):** San Antonio (demand 300), Dallas (demand 900), Houston (demand 800). Total warehouse demand = $2,000\ \text{units/month}$.
- **Balance:** The system is strictly balanced ($\sum \text{Capacity} = \sum \text{Demand} = 2,000$).

Plan 1: Initial Least-Cost Allocation ($Z = \$39,900$)	Plan 2: Alternative Feasible Allocation ($Z = \$42,300$)
	
Plan 1	*Plan 2*

Cost Structure & Allocation Verification:

- **Plan 1 Allocations:**

- Amarillo → Dallas: $400\ \text{units} \times \$21 = \$8,400$
- Waco → San Antonio: $300\ \text{units} \times \$20 = \$6,000$
- Waco → Dallas: $500\ \text{units} \times \$21 = \$10,500$
- Waco → Houston: $200\ \text{units} \times \$30 = \$6,000$
- Huntsville → Houston: $600\ \text{units} \times \$15 = \$9,000$
- **Total Cost:** $\$8,400 + \$6,000 + \$10,500 + \$6,000 + \$9,000 = \mathbf{\$39,900}$ (399×100).

- **Plan 2 Allocations:**

- Amarillo → Dallas: $200\ \text{units} \times \$21 = \$4,200$
- Amarillo → Houston: $200\ \text{units} \times \$42 = \$8,400$
- Waco → San Antonio: $300\ \text{units} \times \$20 = \$6,000$
- Waco → Dallas: $700\ \text{units} \times \$21 = \$14,700$
- Huntsville → Houston: $600\ \text{units} \times \$15 = \$9,000$
- **Total Cost:** $\$4,200 + \$8,400 + \$6,000 + \$14,700 + \$9,000 = \mathbf{\$42,300}$ (423×100).

1.2 Evaluation of Statements (a) through (j)

Statement a)

> *"The first solution with a total cost of $39,900 may be optimal"*

- **Verdict: TRUE**
- **Pedagogical Explanation:** The Least Cost Method is an initial basic feasible solution heuristic. While it does not *guarantee* global optimality by itself, the resulting solution can happen to be the global minimum. Because no feasible solution with a strictly lower cost ($< \$39,900$) has been found, and its optimality has not been mathematically refuted through dual optimality tests (e.g., MODI / Stepping Stone method), it is entirely possible that this solution is optimal. Therefore, stating that it *may* be optimal is correct.

Statement b)

> "*The first solution with a total cost of $39,900 may not be optimal*"

- **Verdict: TRUE**
- **Pedagogical Explanation:** The Least Cost Method operates on a myopic (greedy) heuristic logic: it allocates to the lowest unit cost available without evaluating shadow prices (opportunity costs) of leaving certain cells empty. As demonstrated in Operations Research, greedy initial solutions frequently contain sub-optimal loops that can be improved by shifting flow along stepping-stone cycles. Without computing reduced costs ($c_{ij} - u_i - v_j \geq 0$), we cannot guarantee that an improving loop does not exist. Thus, stating that it *may not* be optimal is correct.

Statement c)

> "*The first solution with a total cost of $39,900 is definitely optimal*"

- **Verdict: FALSE**
- **Pedagogical Explanation:** In formal optimization, a solution can only be declared *definitely optimal* if it satisfies a proven mathematical optimality criterion (such as Kuhn-Tucker conditions, Simplex non-negative reduced costs, or MODI indices $\bar{c}_{ij} \geq 0$ for all non-basic cells). Merely observing that Plan 1 is cheaper than Plan 2 ($39,900 < 42,300$) only proves that Plan 1 is superior to Plan 2—it does not prove that no other feasible solution with a cost lower than $39,900 exists. Therefore, claiming it is *definitely* optimal is scientifically false.

Statement d)

> "*The first solution with a total cost of $39,900 is definitely not optimal*"

- **Verdict: FALSE**
- **Pedagogical Explanation:** To prove that a feasible solution is *definitely not optimal*, one must either demonstrate a specific feasible solution with a lower cost (e.g., $Z < \$39,900$) or prove that the shadow price system yields a negative improvement index ($\bar{c}_{ij} < 0$). Neither condition has been shown here; in fact, $39,900 is the best solution known so far. Claiming it is *definitely not optimal* is unsubstantiated and false.

Statement e)

> "*The second solution with a total cost of $42,300 may be optimal*"

- **Verdict: FALSE**
- **Pedagogical Explanation:** In a cost minimization linear program ($\min Z$), an optimal solution is defined as a feasible solution $x^ $such that Z(x^) \le Z(x) $for all feasible x

$.Since we already have a feasible solution (Plan 1) with a total cost of \$39,900$, and $\$39,900 < \$42,300$, Plan 2 cannot possibly be the minimum-cost solution. Plan 2 is strictly dominated by Plan 1. Therefore, Plan 2 cannot be optimal under any circumstances.

Statement f)

> "*The second solution with a total cost of $42,300 is definitely not optimal*"

- **Verdict: TRUE**
- **Pedagogical Explanation:** Because a known, verified feasible solution exists that achieves a strictly lower total cost ($\$39,900 < \$42,300$), Plan 2 fails the definition of optimality. A candidate solution that is strictly worse than an already discovered feasible plan is conclusively and definitively non-optimal.

Statement g)

> "*An optimal solution of this problem means that any other solutions of the problem will have the same or higher total cost.*"

- **Verdict: TRUE**
- **Pedagogical Explanation:** This is the exact fundamental definition of a global optimum in mathematical programming for minimization problems:

$$Z(x^*) \le Z(x), \quad \forall x \in \mathcal{F}$$

where $\mathcal{F}$ is the feasible region. Any alternative optimal solution will have an identical (same) total cost, and all other non-optimal feasible solutions will have a strictly higher total cost.

Statement h)

> "*The two solutions obtained for this transportation problem are solutions of the "model", may need further evaluations before they are implemented to actually solve the real-world problem.*"

- ***Verdict: TRUE***
- ***Pedagogical Explanation:*** *A mathematical model is an abstraction and simplification of reality. Standard transportation models assume static linear costs, unconstrained road/freight corridors, perfectly reliable carriers, and deterministic demands. In practical industrial engineering, real-world deployment requires evaluating qualitative, operational, and safety factors—such as delivery time windows, highway weight limits, union rules, carrier reliability, contractual volume penalties, and customer service agreements.*

Statement i)

> "*It is possible that a general transportation problem model does not have any solutions, depending on the problem conditions and data such as amounts of supply, demands and other requirements.*"

- ***Verdict: TRUE***
- ***Pedagogical Explanation:*** *In general linear programming and constrained transportation models, infeasibility can arise whenever problem constraints are mutually contradictory. For instance, if certain routes are prohibited ($x_{ij} = 0$), if route capacities are restricted by bottleneck bridges/tunnels ($x_{ij} \le U_{ij}$), if minimum batch sizes are enforced, or if total supply is*

strictly insufficient to satisfy demand without allowing backlog/slack variables, the feasible region $\mathcal{F} = \emptyset$ (no feasible solution exists).

Statement j)

> *"The unit transportation cost data used to find the solutions are likely to be estimated values and careful evaluations are needed to make sure they are accurate."*

- ***Verdict: TRUE***
- ***Pedagogical Explanation:*** *Freight and logistics costs are rarely fixed constants; they are subject to dynamic fuel surcharges, seasonal demand swings, toll adjustments, deadhead mileage, carrier rate negotiations, and varying freight classes. If the input parameters c_{ij} are inaccurate, the resulting solution may be sub-optimal or even counter-productive. Industrial engineers must perform post-optimality sensitivity analysis (evaluating allowable increases and decreases for c_{ij}) to ensure managerial decisions are robust against estimation errors.*

1.3 Problem 1 Master Summary Table

Statement	*Core Concept Analyzed*	*Verdict*	*Key Mathematical / Engineering Justification*
a	*Initial heuristic solution optimality*	***TRUE***	*Feasible heuristic plan can happen to be optimal; optimality is not disproved.*
b	*Heuristic lack of global guarantee*	***TRUE***	*Heuristics do not guarantee global optimality; improving loops may exist.*
c	*Proof of optimality requirement*	***FALSE***	*Lower cost than another plan does not prove global optimality without testing.*
d	*Rejection of candidate plan*	***FALSE***	*Plan 1 is the best solution found; it cannot be declared definitely non-optimal.*
e	*Sub-optimal candidate plan*	***FALSE***	*Plan 2 (*$42, 300) is strictly dominated by Plan 1 (39,900); cannot be optimal.*
f	*Strict non-optimality of dominated plan*	***TRUE***	*Direct counterexample (*$39,900 < 42,300$*) conclusively disproves optimality.*
g	*Definition of global optimality (* $\min Z$*)*	***TRUE***	*By definition, all other feasible solutions have equal or greater cost (Z \ge Z^).*
h	*Model abstraction vs. industrial reality*	***TRUE***	*Real-world factors (time windows, carrier contracts, safety) require validation.*
i	*Feasibility conditions & constrained models*	***TRUE***	*Additional constraints (route caps, prohibited legs) can cause infeasibility.*
j	*Parameter uncertainty & sensitivity analysis*	***TRUE***	*Unit freight costs fluctuate; estimation errors necessitate sensitivity testing.*

Problem 2: Wholesale Center to Store Transportation Problem

2.1 Problem Description & Input Data

*Solve the balanced transportation problem using the **Lowest Cost First (Least Cost Method)** to determine how the supplies from two wholesale distribution centers should satisfy the demands of four regional retail stores at minimum total cost.*

From \ To	Richmond	Philadelphia	Cleveland	Knoxville
Louisville	$90	$90	$100	$70
Morgantown	$85	$70	$60	$130

Store:	Richmond	Philadephia	Cleveland	Knoxville
Demand:	10,000	5,000	5,000	10,000

Wholesale center:	Louisville	Morgantown
Capacity:	10,000	20,000

Problem 2 Data Table

System Parameters:

- ***Wholesale Centers (Supply Sources):***

- *Louisville: Supply* $S_1 = 10,000$ units

- *Morgantown: Supply* $S_2 = 20,000$ units

- ***Total Supply:*** $\sum_{i=1}^{2} S_i = 10,000 + 20,000 = \mathbf{30,000}$ units

- ***Retail Stores (Demand Sinks):***

- *Richmond: Demand* $D_1 = 10,000$ units

- *Philadelphia: Demand* $D_2 = 5,000$ units

- *Cleveland: Demand* $D_3 = 5,000$ units

- *Knoxville: Demand* $D_4 = 10,000$ units

- ***Total Demand:*** $\sum_{j=1}^{4} D_j = 10,000 + 5,000 + 5,000 + 10,000 = \mathbf{30,000}$ units

- ***Balance Condition:*** $\sum S_i = \sum D_j = 30,000$ *(The transportation model is perfectly balanced).*

Unit Transportation Cost Matrix $C = [c_{ij}]$ ***(in*** $\$/\text{unit}$***):***

Wholesale Center \ Retail Store	***Richmond*** (D_1)	***Philadelphia*** (D_2)	***Cleveland*** (D_3)	***Knoxville*** (D_4)	***Capacity*** (S_i)
Louisville	$\$90$	$\$90$	$\$100$	$\$70$	$10,000$
Morgantown	$\$85$	$\$70$	$\$60$	$\$130$	$20,000$
Store Demand (D_j)	$10,000$	$5,000$	$5,000$	$10,000$	$30,000$

2.2 Step-by-Step Execution: Lowest Cost First Method

The Lowest Cost First algorithm greedily allocates the maximum possible quantity to the available cell having the lowest unit transportation cost, updating remaining supplies and demands at each iteration until all requirements are satisfied.

Iteration 1:

- ***Examine all unit costs:***

$$\min\{c_{ij}\} = c_{\text{Morgantown, Cleveland}} = \mathbf{\$60}$$

- ***Evaluate Capacities:***

- Morgantown Capacity: $20,000$

- Cleveland Demand: $5,000$

- ***Allocation:***

$$x_{\text{Morgantown, Cleveland}} = \min(20,000, 5,000) = \mathbf{5,000}\ \text{units}$$

- ***State Updates:***

- Remaining Cleveland Demand: $5,000 - 5,000 = 0$ *(Cleveland is fully satisfied; eliminate Column 3).*

- Remaining Morgantown Capacity: $20,000 - 5,000 = 15,000\ \text{units}$.

Iteration 2:

- ***Examine remaining active cells:***

- Louisville $\rightarrow$ *Richmond:* $\$90$

- Louisville $\rightarrow$ *Philadelphia:* $\$90$

- Louisville $\rightarrow$ *Knoxville:* $\mathbf{\$70}$

- Morgantown $\rightarrow$ *Richmond:* $\$85$

- Morgantown $\rightarrow$ *Philadelphia:* $\mathbf{\$70}$

- Morgantown $\rightarrow$ *Knoxville:* $\$130$

- ***Identify lowest unit cost:*** *There is a tie at* $\mathbf{\$70}$ *between (Louisville, Knoxville) and (Morgantown, Philadelphia).*
- ***Allocation to (Louisville, Knoxville):***

- Louisville Capacity: $10,000$

- Knoxville Demand: $10,000$

-

$$x_{\text{Louisville, Knoxville}} = \min(10,000, 10,000) = \mathbf{10,000} \text{ units}$$

- ***State Updates:***

- Remaining Knoxville Demand: $10,000 - 10,000 = 0$ *(Knoxville is fully satisfied; eliminate Column 4).*

- Remaining Louisville Capacity: $10,000 - 10,000 = 0$ *(Louisville is exhausted; eliminate Row 1).*

Iteration 3:

- ***Examine remaining active cells (only Morgantown row remains):***

- Morgantown $\rightarrow$ Richmond: $\$85$

- Morgantown $\rightarrow$ Philadelphia: $\mathbf{\$70}$

- ***Identify lowest unit cost:*** $c_{\text{Morgantown, Philadelphia}} = \mathbf{\$70}$.
- ***Allocation to (Morgantown, Philadelphia):***

- Remaining Morgantown Capacity: $15,000$

- Philadelphia Demand: $5,000$

-

$$x_{\text{Morgantown, Philadelphia}} = \min(15,000, 5,000) = \mathbf{5,000} \text{ units}$$

- ***State Updates:***

- Remaining Philadelphia Demand: $5,000 - 5,000 = 0$ *(Philadelphia is fully satisfied; eliminate Column 2).*

- Remaining Morgantown Capacity: $15,000 - 5,000 = 10,000$ units.

Iteration 4:

- ***Examine final unsatisfied demand:***

- Only Richmond remains with demand $D_1 = 10,000$ units.

- Only Morgantown has remaining capacity: $10,000$ units.

- Unit Cost: $c_{\text{Morgantown, Richmond}} = \$85$.

- ***Allocation to (Morgantown, Richmond):***

$$x_{\text{Morgantown, Richmond}} = \min(10,000, 10,000) = \mathbf{10,000} \text{ units}$$

- ***State Updates:***

- Remaining Richmond Demand: $10,000 - 10,000 = 0$ *(Satisfied).*

- Remaining Morgantown Capacity: $10,000 - 10,000 = 0$ *(Exhausted).*

(Note on Alternative Tie-Breaker): If (Morgantown, Philadelphia) had been chosen first in Iteration 2, Morgantown's remaining capacity would become $15,000 - 5,000 = 10,000$*; then (Louisville, Knoxville) would receive* $10,000$*, and Richmond would receive* $10,000$ *from Morgantown. The final allocation plan is identical and completely unique.*

2.3 Final Transportation Tableau & Total Cost Calculation

Completed Transportation Tableau:

Wholesale Center	*Richmond ($c = 85/90$)*	*Philadelphia ($c = 70/90$)*	*Cleveland ($c = 60/100$)*	*Knoxville ($c = 70/130$)*	*Total Shipped*	*Center Capacity*
Louisville	—	—	—	***10,000*** *($70)*	***10,000***	*10,000*
Morgantown	***10,000*** *($85)*	***5,000*** *($70)*	***5,000*** *($60)*	—	***20,000***	*20,000*
Total Received	***10,000***	***5,000***	***5,000***	***10,000***	***30,000***	—
Required Demand	*10,000*	*5,000*	*5,000*	*10,000*	—	***30,000***

Total Cost Computation:

$$\text{Total Cost } Z = \sum_{i} \sum_{j} c_{ij} x_{ij}$$

$$Z = (10,000 \times \$70) + (10,000 \times \$85) + (5,000 \times \$70) + (5,000 \times \$60)$$

$$Z = \$700,000 + \$850,000 + \$350,000 + \$300,000$$

$$\mathbf{Z = \$2,200,000}$$

Problem 3: Fork Truck Relocation (Chapter 5, Problem 1)

3.1 Problem Description & Data

A manufacturing plant has fork trucks located at four supply sites that must be redistributed to three demand sites to balance material handling operations across the facility.

Available & Needed Fork Trucks	*Unit Transfer Distance / Cost Matrix (Fig. P5.1)*
1. Fork trucks are located at sites throughout the plant as noted below and are needed at other sites as shown. Formulate as a transportation programming problem and obtain a solution by using the least cost method (see Figure P5.1). Site where fork trucks are available: *Available & Needed*	**Site** A B C D **Number Available** 3 1 2 1 Sites where fork trucks are needed: **Site** X Y Z **Number Needed** 4 2 1 Distance Matrix (Entries in $) From \ To: X Y Z A 5 4 2 B 2 1 2 C 5 4 5 D 3 4 6 **Fig. P5.1** Distance matrix (entries in $). *Distance Matrix*

System Parameters:

- ***Available Sites (Supply):***

- *Site A:* $S_A = 3$ *trucks*

- *Site B:* $S_B = 1$ *truck*

- *Site C:* $S_C = 2$ *trucks*

- *Site D:* $S_D = 1$ *truck*

- ***Total Supply:*** $\sum S_i = 3 + 1 + 2 + 1 = \mathbf{7}$ fork trucks

- ***Needed Sites (Demand):***

- *Site X:* $D_X = 4$ *trucks*

- *Site Y:* $D_Y = 2$ *trucks*

- *Site Z:* $D_Z = 1$ *truck*

- ***Total Demand:*** $\sum D_j = 4 + 2 + 1 = \mathbf{7}$ fork trucks

- ***Balance Condition:*** *Total Available (7) = Total Needed (7).*

Unit Cost Matrix $C = [c_{ij}]$ ***(in $ / truck):***

From \ To	*Site X*	*Site Y*	*Site Z*	*Available (S_i)*
Site A	$5	$4	$2	3
Site B	$2	$1	$2	1
Site C	$5	$4	$5	2
Site D	$3	$4	$6	1
Needed (D_j)	4	2	1	7

3.2 Part 1: Transportation Programming Problem Formulation

1. Decision Variables:

Define x_{ij} as the number of fork trucks transferred from available site i to needed site j:

$$x_{ij} \in \mathbb{Z}_{\geq 0}, \quad \text{for } i \in \{A, B, C, D\} \text{ and } j \in \{X, Y, Z\}$$

2. Objective Function:

Minimize the total transfer cost across all 12 potential relocation routes:

$$\min Z = \sum_{i \in \{A,B,C,D\}} \sum_{j \in \{X,Y,Z\}} c_{ij} x_{ij}$$

$$\begin{aligned}

*\min Z = \;& 5 x_{AX} + 4 x_{AY} + 2 x_{AZ} *

*& + 2 x_{BX} + 1 x_{BY} + 2 x_{BZ} *

*& + 5 x_{CX} + 4 x_{CY} + 5 x_{CZ} *

& + 3 x_{DX} + 4 x_{DY} + 6 x_{DZ}

\end{aligned}$$

3. Constraints:

- ***Supply Constraints (Available Fork Trucks at each site):***

$$x_{AX} + x_{AY} + x_{AZ} = 3 \quad (\text{Site A})$$

$$x_{BX} + x_{BY} + x_{BZ} = 1 \quad (\text{Site B})$$

$$x_{CX} + x_{CY} + x_{CZ} = 2 \quad (\text{Site C})$$

$$x_{DX} + x_{DY} + x_{DZ} = 1 \quad (\text{Site D})$$

- ***Demand Constraints (Needed Fork Trucks at each site):***

$$x_{AX} + x_{BX} + x_{CX} + x_{DX} = 4 \quad (\text{Site X})$$

$$x_{AY} + x_{BY} + x_{CY} + x_{DY} = 2 \quad (\text{Site Y})$$

$$x_{AZ} + x_{BZ} + x_{CZ} + x_{DZ} = 1 \quad (\text{Site Z})$$

- ***Integrity & Non-Negativity:***

$$x_{ij} \in \{0, 1, 2, 3, 4\}, \quad \forall i \in \{A, B, C, D\},\ j \in \{X, Y, Z\}$$

3.3 Part 2: Least Cost Method Solution Execution

Step-by-Step Allocation Procedure:

1. ***Step 1 (Lowest Cost in Table):***

- The minimum unit cost across the entire matrix is $c_{BY} = \mathbf{\$1}$.

- Available at B: 1*; Needed at Y:* 2.

- Allocate: $x_{BY} = \min(1, 2) = \mathbf{1}$ truck.

- Site B is exhausted ($1 - 1 = 0$*; eliminate Row B).*

- Remaining demand at Y: $2 - 1 = 1$ truck.

2. ***Step 2 (Next Lowest Active Unit Cost):***

- Active cells with unit cost $\$2$*: cell* (A, Z) *has* $c_{AZ} = \mathbf{\$2}$*. (Cells* (B, X) *and* (B, Z) *are in row B which is exhausted).*

- Available at A: 3*; Needed at Z:* 1.

- Allocate: $x_{AZ} = \min(3, 1) = \mathbf{1}$ truck.

- Site Z is fully satisfied ($1 - 1 = 0$*; eliminate Column Z).*

- Remaining supply at A: $3 - 1 = 2$ trucks.

3. ***Step 3 (Next Lowest Active Unit Cost):***

- Active remaining cells:

- Row A: $(A, X) = \$5, (A, Y) = \$4$

- Row C: $(C, X) = \$5$*,(C, Y) = \4*

- Row D: $(D, X) = \mathbf{\$3}$*,(D, Y) = \4*

- The minimum active cost is $c_{DX} = \mathbf{\$3}$.

- Available at D: 1*; Needed at X:* 4.

- Allocate: $x_{DX} = \min(1, 4) = \mathbf{1}$ truck.

- Site D is exhausted ($1 - 1 = 0$*; eliminate Row D).*

- Remaining demand at X: $4 - 1 = 3$ trucks.

4. ***Step 4 (Resolving the Cost*** $\$4$ ***Tie):***

- Active remaining cells: $(A, X) = \$5, (A, Y) = \mathbf{\$4}, (C, X) = \$5, (C, Y) = \mathbf{\$4}$.

- Both (A, Y) *and* (C, Y) *have unit cost* $\$4$*. We evaluate both tie-breaking paths:*

- ***Path 1 (Allocate to*** $A \to Y$ ***first):***

- Remaining demand at Y: 1*; Available at A:* 2.

- Allocate: $x_{AY} = \min(2, 1) = \mathbf{1}$ truck.

- Site Y is fully satisfied ($1 - 1 = 0$*; eliminate Column Y).*

- Remaining supply at A: $2 - 1 = 1$ truck.

- Only Site X remains with unsatisfied demand = 3*:*

- Allocate remaining supply of A: $x_{AX} = \min(1, 3) = \mathbf{1}$ truck *(Site A exhausted).*

- Allocate remaining supply of C: $x_{CX} = \min(2, 2) = \mathbf{2}$ trucks *(Site C exhausted, Site X satisfied).*

- ***Cost for Path 1:***

$$Z_1 = (1 \times 1) + (1 \times 2) + (1 \times 3) + (1 \times 4) + (1 \times 5) + (2 \times 5) = 1 + 2 + 3 + 4 + 5 + 10 = \mathbf{\$25}$$

- ***Path 2 (Allocate to $C \to Y$ first):***

- *Remaining demand at Y: 1; Available at C: 2.*

- *Allocate: $x_{CY} = \min(2, 1) = \mathbf{1}$ truck.*

- *Site Y is fully satisfied; remaining supply at C: $2 - 1 = 1$ truck.*

- *Only Site X remains with unsatisfied demand = 3:*

- *Allocate remaining supply of A: $x_{AX} = \min(2, 3) = \mathbf{2}$ trucks (Site A exhausted).*

- *Allocate remaining supply of C: $x_{CX} = \min(1, 1) = \mathbf{1}$ truck (Site C exhausted, Site X satisfied).*

- ***Cost for Path 2:***

$$Z_2 = (1 \times 1) + (1 \times 2) + (1 \times 3) + (1 \times 4) + (2 \times 5) + (1 \times 5) = 1 + 2 + 3 + 4 + 10 + 5 = \mathbf{\$25}$$

Conclusion: *Both tie-breaking conventions yield the identical minimum relocation cost of **$25**.*

3.4 Final Transportation Tableaux

Primary Solution Tableau (Path 1):

Available Site	*Needed Site X (c)*	*Needed Site Y (c)*	*Needed Site Z (c)*	*Total Shipped*	*Available*
Site A	***1*** *($5)*	***1*** *($4)*	***1*** *($2)*	***3***	*3*
Site B	—	***1*** *($1)*	—	***1***	*1*
Site C	***2*** *($5)*	—	—	***2***	*2*
Site D	***1*** *($3)*	—	—	***1***	*1*
Total Received	***4***	***2***	***1***	***7***	—
Required Demand	*4*	*2*	*1*	—	***7***

$$\mathbf{Z = (1 \times 5) + (1 \times 4) + (1 \times 2) + (1 \times 1) + (2 \times 5) + (1 \times 3) = \$25}$$

Alternate Solution Tableau (Path 2):

Available Site	*Needed Site X (c)*	*Needed Site Y (c)*	*Needed Site Z (c)*	*Total Shipped*	*Available*
Site A	***2*** *($5)*	—	***1*** *($2)*	***3***	*3*
Site B	—	***1*** *($1)*	—	***1***	*1*
Site C	***1*** *($5)*	***1*** *($4)*	—	***2***	*2*
Site D	***1*** *($3)*	—	—	***1***	*1*
Total Received	***4***	***2***	***1***	***7***	—
Required Demand	*4*	*2*	*1*	—	***7***

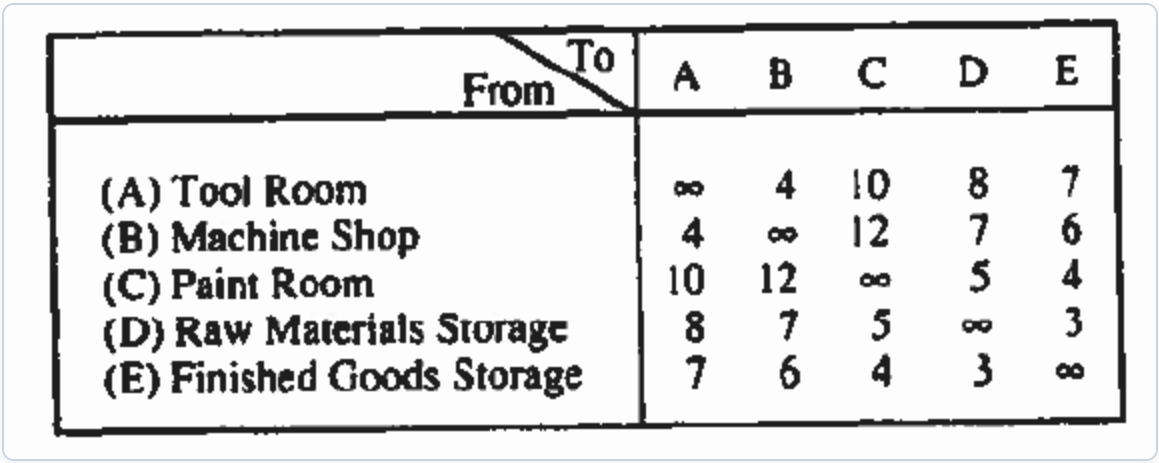
$$\mathbf{Z} = (\mathbf{2} \times \mathbf{5}) + (\mathbf{1} \times \mathbf{2}) + (\mathbf{1} \times \mathbf{1}) + (\mathbf{1} \times \mathbf{5}) + (\mathbf{1} \times \mathbf{4}) + (\mathbf{1} \times \mathbf{3}) = \mathbf{\$25}$$

Problem 4: Traveling Salesman Problem (TSP) Fork Truck Routing

4.1 Problem Description & Distance Matrix

*A warehouse forklift truck is currently located at the **Tool Room (Node A)** and must visit four other departments before returning to base:*

- ***(A)** Tool Room (Starting / Home Depot)*
- ***(B)** Machine Shop*
- ***(C)** Paint Room*
- ***(D)** Raw Materials Storage*
- ***(E)** Finished Goods Storage*



From \ To	A	B	C	D	E
(A) Tool Room	∞	4	10	8	7
(B) Machine Shop	4	∞	12	7	6
(C) Paint Room	10	12	∞	5	4
(D) Raw Materials Storage	8	7	5	∞	3
(E) Finished Goods Storage	7	6	4	3	∞

Problem 4 Distance Matrix

***Symmetric Distance Matrix** $D = [d_{ij}]$:*

From \ To	*(A) Tool Room*	*(B) Machine Shop*	*(C) Paint Room*	*(D) Raw Materials*	*(E) Finished Goods*
(A) Tool Room	∞	4	10	8	7
(B) Machine Shop	4	∞	12	7	6
(C) Paint Room	10	12	∞	5	4
(D) Raw Materials Storage	8	7	5	∞	3
(E) Finished Goods Storage	7	6	4	3	∞

4.2 Part 1: Formal Traveling Salesman Problem (TSP) Formulation

1. Network & Decision Variables:

Let $G = (V, E)$ be a complete undirected graph where vertices $V = \{A, B, C, D, E\}$ ($n = 5$).

Define binary routing variables:

$$x_{ij} = \begin{cases} 1, & \text{if the forklift travels directly from location } i \text{ to location } j \\ 0, & \text{otherwise} \end{cases} \qquad \forall i, j \in V,\ i \neq j$$

2. Objective Function:

Minimize the total tour distance:

$$\min Z = \sum_{i \in V} \sum_{j \in V, j \neq i} d_{ij} x_{ij}$$

3. Degree / Assignment Constraints:

Every facility must be departed exactly once and entered exactly once:

$$\sum_{j \in V, j \neq i} x_{ij} = 1, \quad \forall i \in V \quad (\text{Out-degree constraint})$$

$$\sum_{i \in V, i \neq j} x_{ij} = 1, \quad \forall j \in V \quad (\text{In-degree constraint})$$

4. Miller-Tucker-Zemlin (MTZ) Subtour Elimination Constraints:

To guarantee that the solution forms a single closed Hamiltonian cycle encompassing all 5 nodes rather than disconnected smaller sub-loops (such as $A \leftrightarrow B$ and $C \leftrightarrow D \leftrightarrow E$):

Let auxiliary continuous sequence variables $u_i \in \mathbb{R}$ represent the visit order of location i (setting $u_A = 1$ for the home depot):

$$u_i - u_j + n x_{ij} \leq n - 1, \quad \forall i, j \in \{B, C, D, E\},\ i \neq j$$

$$1 \leq u_i \leq n - 1, \quad \forall i \in \{B, C, D, E\}$$

$$x_{ij} \in \{0, 1\}, \quad \forall i, j \in V$$

4.3 Part 2: Nearest-Neighbor (NN) Method Solution

The Nearest-Neighbor heuristic starts at Node A and iteratively moves to the closest unvisited node.

```
Start at Depot A:
  Distances from A: B(4), E(7), D(8), C(10)
  → Select closest: B (distance 4)
  Tour: A → B

From Node B:
  Unvisited: {C, D, E}
  Distances from B: E(6), D(7), C(12)
  → Select closest: E (distance 6)
  Tour: A → B → E

From Node E:
  Unvisited: {C, D}
  Distances from E: D(3), C(4)
  → Select closest: D (distance 3)
  Tour: A → B → E → D

From Node D:
  Unvisited: {C}
  Only 1 unvisited node remains: C
  → Move to: C (distance 5)
  Tour: A → B → E → D → C

Return to Home Depot A:
  From last node C, return to start A:
  → Distance C → A = 10
  Completed Cycle: A → B → E → D → C → A
```

Nearest-Neighbor Distance Summary:

Leg #	*Route Segment*	*Departure Node*	*Arrival Node*	*Segment Distance*	*Cumulative Distance*
1	*Outbound 1*	*(A) Tool Room*	*(B) Machine Shop*	4	4
2	*Outbound 2*	*(B) Machine Shop*	*(E) Finished Goods*	6	10
3	*Outbound 3*	*(E) Finished Goods*	*(D) Raw Materials*	3	13
4	*Outbound 4*	*(D) Raw Materials*	*(C) Paint Room*	5	18
5	*Return Leg*	*(C) Paint Room*	*(A) Tool Room*	10	28

- ***Nearest-Neighbor Tour Sequence:*** $\mathbf{A \rightarrow B \rightarrow E \rightarrow D \rightarrow C \rightarrow A}$
- ***Total Round-Trip Travel Distance:*** $4 + 6 + 3 + 5 + 10 = \mathbf{28}$
- *(One-way path without return to A:* $4 + 6 + 3 + 5 = 18$*).*

4.4 Part 3: "Next-Nearest-Neighbor" (NNN) Method Solution

The assignment specifies:

> *"In using this method, each time the truck will go to the next stop which is the next-nearest neighbor instead of the nearest neighbor of its current position. Calculate the total travel distance of this solution."*

We evaluate both the strict literal interpretation and the classroom/midterm variant:

Method A: Strict Literal Rule ("Each Time" Select 2nd-Nearest Unvisited Node)

```
Start at Depot A:
  Unvisited: {B(4), E(7), D(8), C(10)}
  1st Nearest: B (4)
  2nd Nearest (Next-Nearest): E (7)
  → Select: E (distance 7)
  Tour: A → E

From Node E:
  Unvisited: {B, C, D}
  Distances: D(3), C(4), B(6)
  1st Nearest: D (3)
  2nd Nearest (Next-Nearest): C (4)
  → Select: C (distance 4)
  Tour: A → E → C

From Node C:
  Unvisited: {B, D}
  Distances: D(5), B(12)
  1st Nearest: D (5)
  2nd Nearest (Next-Nearest): B (12)
  → Select: B (distance 12)
  Tour: A → E → C → B

From Node B:
  Unvisited: {D}
  Only 1 node remains; must visit: D (distance 7)
  Tour: A → E → C → B → D

Return to Home Depot A:
  From last node D, return to start A:
  → Distance D → A = 8
  Completed Cycle: A → E → C → B → D → A
```

Method A Distance Summary:

Leg #	*Route Segment*	*Departure Node*	*Arrival Node*	*Selection Rule Applied*	*Segment Distance*
1	*Outbound 1*	*(A) Tool Room*	*(E) Finished Goods*	*2nd nearest among {B, C, D, E}*	7
2	*Outbound 2*	*(E) Finished Goods*	*(C) Paint Room*	*2nd nearest among {B, C, D}*	4
3	*Outbound 3*	*(C) Paint Room*	*(B) Machine Shop*	*2nd nearest among {B, D}*	12
4	*Outbound 4*	*(B) Machine Shop*	*(D) Raw Materials*	*Last remaining unvisited node*	7
5	*Return Leg*	*(D) Raw Materials*	*(A) Tool Room*	*Return to home base A*	8

- ***Method A Tour Sequence:*** $\mathbf{A \rightarrow E \rightarrow C \rightarrow B \rightarrow D \rightarrow A}$
- ***Total Round-Trip Travel Distance:*** $7 + 4 + 12 + 7 + 8 = \mathbf{38}$
- *(One-way path without return to A:* $7 + 4 + 12 + 7 = 30$*).*

Method B: Concordia Classroom / Midterm Variant (2nd-Nearest on Leg 1, Nearest Neighbor Thereafter)

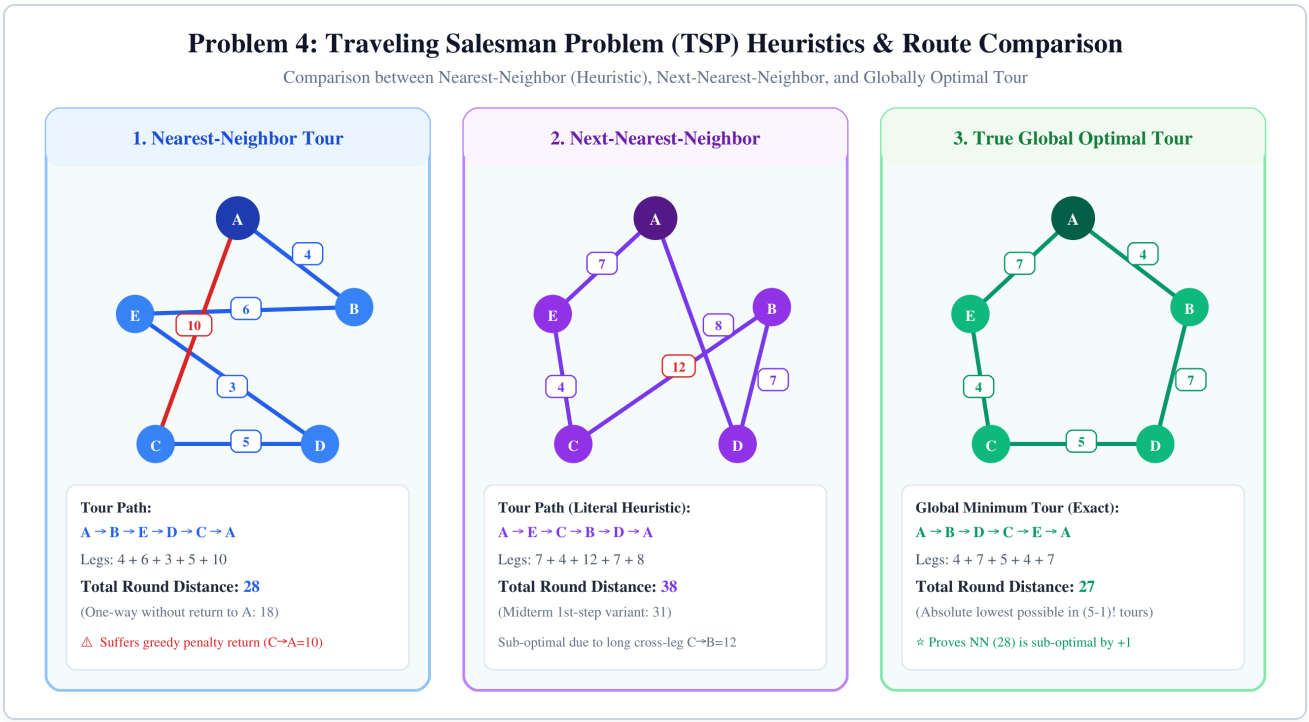
As documented in the official INDU 211 Midterm Master Quantitative Guide:

> *"Select the second-nearest neighbor on the first leg out of depot A, then revert strictly to the nearest-neighbor rule for all subsequent legs."*

1. ***Step 1 (Leg out of A):*** *2nd nearest is* ***E*** *($d_{AE} = 7$). Visited: $\{A, E\}$. Remaining: $\{B, C, D\}$.*
2. ***Step 2 (From E):*** *Closest unvisited node is* ***D*** *($d_{ED} = 3$). Visited: $\{A, E, D\}$. Remaining: $\{B, C\}$.*
3. ***Step 3 (From D):*** *Distances: $C(5)$ vs $B(7)$. Closest is* ***C*** *($d_{DC} = 5$). Visited: $\{A, E, D, C\}$. Remaining: $\{B\}$.*
4. ***Step 4 (From C):*** *Only remaining node is* ***B*** *($d_{CB} = 12$). Visited: all.*
5. ***Step 5 (Return from B to A):*** *Return distance $d_{BA} = 4$.*

- ***Method B Tour Sequence:*** $\mathbf{A \to E \to D \to C \to B \to A}$
- ***Total Round-Trip Travel Distance:*** $7 + 3 + 5 + 12 + 4 = \mathbf{31}$
- *(One-way path without return to A: $7 + 3 + 5 + 12 = 27$).*

4.5 Pedagogical Comparative Analysis: The "Greedy Trap" & Global Optimal Tour



TSP Routing Comparison

Combinatorial Verification of All $(5-1)!/2 = 12$ Unique Tours:

To understand whether either heuristic found the true optimal tour, we evaluate all 12 distinct undirected Hamiltonian cycles originating and terminating at Tool Room A:

Tour Permutation	*Route Sequence*	*Leg Distances*	*Total Round-Trip Distance*	*Status*
1	$A \to B \to C \to D \to E \to A$	$4 + 12 + 5 + 3 + 7$	31	*Feasible*
2	$A \to B \to C \to E \to D \to A$	$4 + 12 + 4 + 3 + 8$	31	*Feasible*
3	$\mathbf{A \to B \to D \to C \to E \to A}$	$\mathbf{4 + 7 + 5 + 4 + 7}$	$\mathbf{27}$	***★ TRUE GLOBAL OPTIMAL TOUR ★***
4	$A \to B \to D \to E \to C \to A$	$4 + 7 + 3 + 4 + 10$	28	*Feasible*
5	$\mathbf{A \to B \to E \to D \to C \to A}$	$\mathbf{4 + 6 + 3 + 5 + 10}$	$\mathbf{28}$	***Nearest-Neighbor Tour (Sub-optimal)***
6	$A \to B \to E \to C \to D \to A$	$4 + 6 + 4 + 5 + 8$	27	*(Mirror of Tour 3)*
7	$A \to C \to B \to D \to E \to A$	$10 + 12 + 7 + 3 + 7$	39	*Feasible*
8	$A \to C \to D \to B \to E \to A$	$10 + 5 + 7 + 6 + 7$	35	*Feasible*
9	$A \to C \to E \to B \to D \to A$	$10 + 4 + 6 + 7 + 8$	35	*Feasible*
10	$A \to D \to B \to C \to E \to A$	$8 + 7 + 12 + 4 + 7$	38	*Feasible*
11	$\mathbf{A \to E \to C \to B \to D \to A}$	$\mathbf{7 + 4 + 12 + 7 + 8}$	$\mathbf{38}$	***Next-Nearest Tour (Method A)***
12	$\mathbf{A \to E \to D \to C \to B \to A}$	$\mathbf{7 + 3 + 5 + 12 + 4}$	$\mathbf{31}$	***Next-Nearest Tour (Method B)***

Key Industrial Engineering Insights:

1. ***The "Greedy Trap" of Nearest Neighbor:***

At Node B, the Nearest-Neighbor heuristic greedily chooses to visit E ($d_{BE} = 6$) instead of D ($d_{BD} = 7$), saving 1 unit locally. However, this early greed isolates Node C at the end of the tour, forcing a severe penalty leg of $d_{CA} = 10$ to return to base. By contrast, the global optimal tour accepts a slightly longer leg ($B \to D = 7$) early on, which unlocks the smooth return $C \to E \to A$ ($4 + 7 = 11$), achieving an overall shorter total distance of 27 vs. 28.

2. ***Computational Complexity:***

For $n = 5$, exhaustive search evaluates $4!/2 = 12$ tours instantaneously. However, as n scales to 20 or 50 warehouse picking stops, the search space grows as $(n-1)!$, which is computationally intractable (NP-hard). This demonstrates why industrial engineers utilize integer programming branch-and-bound solvers (such as CPLEX or Gurobi) or meta-heuristics (Genetic Algorithms, 2-Opt local search) for industrial routing.

Final Results Summary Sheet

Problem	*Topic / Method*	*Key Output / Metric*	*Final Numerical Result*
Problem 1	*Conceptual Diagnostic (Plain View Transportation Model)*	*Statements (a) through (j)*	***a:*** *TRUE,* ***b:*** *TRUE,* ***c:*** *FALSE,* ***d:*** *FALSE,* ***e:*** *FALSE,* ***f:*** *TRUE,* ***g:*** *TRUE,* ***h:*** *TRUE,* ***i:*** *TRUE,* ***j:*** *TRUE*
Problem 2	*Lowest Cost First Distribution Allocation*	*Total Transportation Cost (Z)*	$\mathbf{\$2,200,000}$ *(Louisville $\rightarrow$ Knoxville: 10k; Morgantown $\rightarrow$ Richmond: 10k, Phila: 5k, Cleve: 5k)*
Problem 3	*LP Formulation & Least Cost Method (Fork Trucks)*	*Total Relocation Cost (Z)*	$\mathbf{\$25}$ *(Allocations: $x_{BY}=1, x_{AZ}=1, x_{DX}=1, x_{AY}=1, x_{AX}=1, x_{CX}=2$)*
Problem 4.1	*Nearest-Neighbor Heuristic (Forklift TSP)*	*Route Sequence & Round Distance*	$\mathbf{A \rightarrow B \rightarrow E \rightarrow D \rightarrow C \rightarrow A}$ *Total Distance =* 28 *(One-way =* 18*)*
Problem 4.2	*Next-Nearest-Neighbor Heuristic (Literal / Variant)*	*Route Sequence & Round Distance*	***Literal (Each Time):*** $\mathbf{A \rightarrow E \rightarrow C \rightarrow B \rightarrow D \rightarrow A}$ *(*38*)* ***Midterm Variant:*** $\mathbf{A \rightarrow E \rightarrow D \rightarrow C \rightarrow B \rightarrow A}$ *(*31*)*
Problem 4.3	*Global Optimal TSP Tour (Exhaustive Benchmark)*	*Optimal Sequence & Min Distance*	$\mathbf{A \rightarrow B \rightarrow D \rightarrow C \rightarrow E \rightarrow A}$ *Minimum Distance =* 27 *(Proves NN is sub-optimal by* $+1$*)*
